Squad Swarm Operating Procedure

SOP Card

8-Robot Heterogeneous Swarm — For Operator Use

문서번호 SAN-OPS-SOP-001

v1.1

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Security	Internal use

Revision History

Version	Changes
v1.0	Initial release. Core rules + presets + modes + Tier colors + emergency procedures.
v1.1	Section 7 FAQ added (surveillance distribution, video viewing, Hub UGV dual SBC).
v1.2	PM decisions D1~D7 + A1~A5 applied: (D1) 8-robot squadron (Leader S1 + Hub S2 + F1-F6 = S3-S8); (A1) Core rule #4 updated — production/demo/**bench** yaml overlay modes; (A2) Section 8 Demo procedure (DEMO 6-phase) added; (A3) command_id echo pattern guidance; (A4) Node watchdog 3s policy.
v1.4	Deputy UGV + Limp Mode operational guidance: (1) Section 7 FAQ — Deputy UGV role, Leader succession priorities; (2) Section 7 FAQ — Limp Mode operational restrictions (no fire, RTB recommended); (3) Squadron composition updated (4~8 variable).

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1. Core Operational Rules (Mandatory Knowledge)

#	Rule
1	On "Assault" command, d = 15m applied automatically — no threat level judgment required
2	Leader (Go2) auto-limited to 1.3 m/s — never command higher speed or overtaking
3	Near cliffs/ravines, auto-transition to single column — override available within 5s
4	deployment_mode yaml overlay 3 types: production(live) / demo(show, weapons disabled) / bench(internal PC test) — bench/demo never in live operations
5	Fire authorization (Fire Confirm) requires operator touch input — no auto-fire
6	Leader battery < 30% triggers Hub UGV auto-takeover — mission continues
7	v1.1 NEW: Camera surveillance area auto-assigned by Leader every 10s — no operator action needed
8	v1.1 NEW: View Video by tapping robot icon on map → "View Video" (Pull mode)

2. Operational Presets (Android APP → Formation Select)

Preset	d	θ	Use	Selection Criterion
Narrow Penetration	3 m	40°	Narrow corridor, woods, alley	Passage width < 6m
Recon/Defense	5 m	90°	360° viewport, flank perimeter	Recon, patrol, base defense
Wide Recon	7 m	120°	Distributed surveillance	Open terrain wide recon
Assault	15 m	60°	Forward firepower + shell evasion	Enemy contact / engagement / breakthrough

3. Operating Modes (Android APP → Mode Select)

Mode	Auto Flow	Operator Action
Offensive	Assembly → Column → V → Assault → Rank → Merge → Column	"Assault" button tap, "Confirm Fire" on engagement
Defensive	Diamond → Drone detection → Auto share → Neutralize	"Defense" button tap, position designate then monitor
Return (RTH)	Auto convert to single column, return to home	"Return" button tap or auto (battery threshold)

4. Follower Status Colors (Map Screen)

Color	Tier	Meaning	Operator Action
Green	T0 Predictive Track	Normal (1s prediction position tracking)	None (normal)
Cyan	T1.5 Auto Reroute	Local obstacle ±2m detour in progress	None (auto-recovery)
Yellow-Green	T1 Normal	Body-frame offset tracking (no prediction)	Check comm status
Yellow	T2 Catch-Up	Lagging, 1.2x speed	Consider lowering Leader speed
Orange	T3 Hard Catch-Up	Max speed pursuit	Consider pause, check comm
Red	T4 Breadcrumb (blinking)	Returning via Leader past path — DANGER	Halt, situation review, RTH if needed

5. Emergency Procedures

5.1 Communication Loss (No Leader Response)

1. Within 30s: Followers self-handle via Tier 1/2/3. No action needed.
2. > 30s: Auto T4 Breadcrumb entry. All followers display blinking.
3. > 60s: Auto RTH (return to home). "Cancel RTH" button overrides.
4. LoRa emergency channel allows emergency commands (stop, return).

5.2 Drone Threat Detection (RF Scanner Alert)

5. Auto alert screen + drone bearing/distance display
6. Within 1km: "Jam" button → Soft-kill auto (operator confirm)
7. Within 150m: "Confirm Fire" button → Hard-kill firing (gyro stabilized)

5.3 Enemy Personnel Detection (AI Object Detection)

8. "Potential Threat" yellow marker + distance/bearing auto-calculation
9. Target tap → wind/trajectory-adjusted "Predicted Hit Point" display
10. "Confirm Fire" → gyro stabilized + precision strike (within 300m)

5.4 Leader Loss (Battery / Damage / Comm Loss)

11. Battery < 30%: Hub UGV auto-takeover (5s alert)
12. Damage/full comm loss: Modified Raft priority succession (Hub UGV 1st)
13. Reform time approximately 5-8s. "Re-organizing" displayed.
14. On new leader decision, mission auto-continues.

5.5 GPS Jamming (RTK Fix Loss)

15. 0-5s: Dead-reckoning short-term hold (formation stable)
16. 5-30s: LiDAR SLAM augmentation (Robosense E1)
17. > 30s: Tier 4 autonomous mode. Return to safe area.
18. Operator: acknowledge "GPS Loss" warning, prefer halt command

6. Quick Decision Reference

Situation	Recommended Action
Mountain/valley march	Auto single column. Accept (or no action)
Urban alley passage	Select "Narrow Penetration" preset (d=3m)
Pre-contact (need visibility)	"Recon/Defense" (d=5m, θ=90°) or "Wide Recon" (d=7m, θ=120°)
Enemy ID → Assault order	"Assault" button (auto d=15m, no threat judgment)
Base 360° perimeter	"Defensive Mode" + "Recon/Defense" preset (auto diamond)
Multi-drone simultaneous intrusion	Auto free-spread entry, verify "Auto Share" ON
Low battery alert	"Return (RTH)" button (auto return to home)

7. v1.1 FAQ — Camera / Video / SLAM

7.1 Q: How is camera surveillance area determined?

A: The Leader robot auto-distributes sectors every 10s. No operator action required.

Robot	ID	Coverage	Operation
Leader (Go2)	S1	Front 0° ±30°	Fixed (camera 1x Wide, HFOV 60°)
Hub UGV	S2	Rear ±90° (180°)	Pan-tilt sweep 30°/sec (8s period)
F1 / F2	S3/S4	Front-left/right 50° each	Pan-tilt sweep 30°/sec
F3 / F4	S5/S6	Left/right flank 50° each	Pan-tilt sweep
F5 / F6	S7/S8	Left/right rear 50° each	Pan-tilt sweep

On threat detection, the Leader immediately reassigns sectors to focus followers on the threat direction. On follower loss, remaining followers auto-redistribute.

7.2 Q: I want to view video from a specific robot.

A: Tap the robot icon on the map → select "View Video" menu (Pull mode).

19. Tap robot icon on map
20. Select "View Video" from popup menu
21. Quality selection: High (HD) / Standard (FHD, default) / Low / Thumbnail
22. Hub UGV starts GStreamer SRT pipeline (display within 200ms)
23. To end: Tap "Stop View Video" → auto cleanup

Multiple videos: If 4+ requested simultaneously, auto-downgrade to "Thumbnail" mode (320x240). Up to 9 robot videos monitorable simultaneously.

7.3 Q: "Hub UGV Comm SBC Failure" alert appeared.

A: The Hub UGV has dual SBC architecture, allowing partial operation when one fails.

Alert	Cause	Mission Impact	Action
"SLAM SBC Failure"	Hub SBC #1 failure	Global map update suspended, local SLAM OK	Mission continues, inspect after return
"Comm SBC Failure"	Hub SBC #2 failure	LTE/video stream suspended, mesh	Video viewing unavailable, mission

Alert	Cause	Mission Impact	Action
		OK	continues
"Both SBCs Failure"	Both SBCs failed (rare)	Leader succession risk	Immediate RTH consideration

7.4 Q: Global map refreshes every 30-60s — is this normal?

A: From v1.1, SLAM aggregated bundle refreshes every 30s (default). Rationale for not 1-second real-time refresh:

- Real-time avoidance handled by each robot's local SLAM at 1Hz (immediate avoidance)
- Global map is for mission planning/recon — 30s refresh sufficient
- 100x comm load reduction frees bandwidth for 1s prediction broadcast

PM may adjust period based on operation (urban: 15s, wide recon: 60s).

7.5 Q: What LiDAR models are used?

Robot	ID	LiDAR Model	Characteristics
Leader (Go2)	S1	Unitree L1 (built-in)	360° rotating, 30m range
Hub UGV	S2	Robosense E1 (solid-state)	200m range, 120° x 90° FoV (v1.3 correction)
Follower UGV	S3-S8	Robosense E1 (solid-state)	200m range, 120° FoV (6 units)

Robosense E1 is a solid-state LiDAR with no moving parts, robust against military vibration/shock. Capable of long-range 200m reconnaissance and effective when combined with RF scanner cross-validation.

7.6 Q: How does it operate in weak Wi-Fi environments?

A: OpenWrt 24.10 Wi-Fi 6 Mesh maintains connectivity via multi-path.

- Mesh node auto-routing (batman-adv BATMAN V)
- AP steering DAWN auto-selects optimal AP
- On Wi-Fi loss, mwan3 auto-switches to LTE backhaul within 8s
- Last resort: LoRa emergency channel (stop, return commands only)

7.7 Q: If enemy detected during sweep?

A: Auto-transition to Track mode, Pan-tilt locks on target. Operator action:

24. "Potential Threat" yellow marker auto-displayed
25. Leader auto-reassigns nearby 2-3 followers' sectors to threat direction (cross-cueing)

26. Operator: tap to confirm target
27. Select Confirm Fire or "Observe Only"

7.8 Q: How is automatic avoidance determined? (v1.3 new)

A: Each UGV computes a Local Cost Map (obstacle map) on its own SBC every second. If hazard cells exist on the predicted path, Tier 1.5 is triggered and an automatic +/-2m lateral detour is performed.

Condition	Cost Map class	Action
Obstacle height >= 235 mm	lethal (cost 254)	**Mandatory detour** (+/-2m lateral)
Ditch width >= 220 mm	lethal	Mandatory detour
Slope > 30 deg	lethal	Mandatory detour
Obstacle 100-235 mm	inflated (cost 200)	Detour if possible, slow pass
Slope 15-30 deg	cautious (cost 100)	Slow pass
Slope < 15 deg, obstacle < 100 mm	free (cost 0)	Normal pass

After **3 consecutive avoidance failures**, an alert "Manual intervention required" appears on the operator screen.

7.9 Q: "LTE Backup Activated" alert appeared (v1.3 new)

A: When Hub UGV (S2) LTE modem is in a non-operational state, Follower S3 automatically takes over the LTE gateway role. The mission continues, and telemetry remains normal (up to 30 sec temporary gap is possible).

Status Display	Meaning	Recommended Action
"Hub LTE Normal" (Green)	Hub S2 operating LTE gateway	Normal — no action
"LTE Backup Active — S3" (Yellow)	S3 took over LTE backup (Hub LTE down)	Continue mission, inspect Hub LTE after return
"LTE Total Outage" (Red)	Hub + S3 LTE both non-operational	Operate on Wi-Fi 6 mesh only, approach operator
"Hub LTE Recovered — S3 demoted" (Green)	Hub LTE recovered, normal distribution	Normal — verify split-brain prevented

On LTE backup follower (S3) failure, auto-chain promotion to next candidate (S5). Backup activation threshold time <= 10s.

7.10 Q: Is the SLAM global map updated every 5 seconds? (v1.3 updated)

A: From v1.3, the Hub UGV SLAM aggregation period is shortened from **30-60s -> 5 sec**. This is to maintain consistency with the Cost Map (1Hz). The operator screen's global map now refreshes every 5 seconds.

- Real-time avoidance uses Local SLAM + Cost Map (1Hz) — instant action
- Global map reconnaissance info refreshes at 5-second intervals
- Can be shortened to 3 sec by PM decision for urban/indoor penetration

8. Demo Procedure (DEMO 6-Phase Sequencer)

8.1 Demo Mode Entry Conditions

- deployment_mode = `demo` (yaml overlay applied)
- Weapons auto-disabled (gun_trigger_permission = DENIED forced)
- BB pellets / blanks or dummy mode operation
- Operator taps "Start Demo" button to initiate auto-sequence

8.2 DEMO 6-Phase Auto Sequence

Phase	Action	Default Params	Operator
DEMO_PHASE_IDLE	Wait for sequence trigger	—	Tap Start Demo
DEMO_PHASE_FORWARD	Advance demo_forward_distance_m	Default 5 m, 0.5 m/s	Observe
DEMO_PHASE_SCAN	Auto scan specified pan/tilt range	pan ±60°, tilt -10° to +5°	Auto advance on target detect
DEMO_PHASE_ENGAGE	Lock target → fire demo_fire_duration_sec	Default 2 sec, BB/dummy	Observe (auto-fire alert)
DEMO_PHASE_REVERSE	Reverse demo_reverse_distance_m	Default 5 m, 0.5 m/s	Observe
DEMO_PHASE_COMPLETE	Return to IDLE, show result screen	—	Tap Confirm

8.3 Forced Stop / Resume

28. Top-right "Stop Demo" button → immediate IDLE return (resetDemoSequence)
29. EmergencyStop input immediately halts SCAN/ENGAGE phase
30. After stop: "Restart Demo" available (begins again from FORWARD)

8.4 Recommended Demo Environment

- Indoor demo: Narrow Penetration preset (d=3m), Leader Go2 + Hub UGV + 2-3 followers
- Outdoor (small venue): Recon/Defense preset (d=5m), Leader + Hub + 4 followers
- DAPA/VIP formal demo: Full 8 robots, Wide Recon preset

8.5 Operator Demo Display

- Current Phase displayed in large text (FORWARD / SCAN / ENGAGE / REVERSE)

- 8-robot positions + sectors + target markers in real time
- "Demo Mode" label persistently displayed (yellow banner)
- `deployment_mode=demo, weapons disabled` status displayed

9. Appendix

9.1 Color Guide (Map Markers)

Color	Meaning
Green	Normal follower / surveillance area
Blue	Leader (Go2) / protected area
Purple	Hub UGV / comm relay area
Yellow	Potential threat / Tier 2-3 follower
Red	Confirmed threat / enemy / Tier 4 follower
Rainbow	Multi-sector distribution (rotates every 10s)

9.2 Emergency Contact

On suspected robot loss/capture: "Compromise" button → immediate certificate revocation for that robot. CRL broadcasts to entire swarm, making that robot unusable.

9.3 Related Documents

- SAN-SDD-SWARM-001 v1.4 — Unified SDD (detailed design)
- SAN-SDD-SUR-001 v1.4 — Surveillance/SLAM/Video Supplementary SDD
- SAN-IDS-CMD-001 v1.4 — Interface Definition Specification

─ This SOP is an operational summary of SAN-SDD-SWARM-001 v1.4. Refer to Unified SDD for details ─